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PAPER_1093: SCm Phonon-Modulated CMB Temperature Fluctuation

Abstract

The SCm temperature fluctuation at angle θ on the CMB sky is:

$$\Delta T_{\mathrm{CMB}}(\theta) = T_0 \cdot S_{26}^{(3)}([\mathrm{SSq}]) \cdot \Phi_{1.25\,\mathrm{THz}}(\omega, \Gamma) \cdot (2R - 1)$$

This derives the anisotropy pattern directly from SCm vacuum buoyancy without requiring an inflaton field, dark energy, or ΛCDM tuning. The observed $\Delta T/T \sim 10^{-5}$ emerges from the product of the cubed Ramanujan sum S_{26}^3 and the phonon resonance amplitude Φ.

$\S$1 Core Equation

$$\Delta T_{\mathrm{CMB}}(\theta) = T_0 \cdot S_{26}^3 \cdot \Phi(\Gamma) \cdot (2R - 1)$$

Symbol	Value	Meaning
T_0	2.7255 K	CMB monopole temperature
S_{26}	$\sum_{k=1}^{26} e^{-[\mathrm{SSq}]\,k/26} = 1.8594$	26-layer Ramanujan sum
S_{26}^3	6.4333	Cubed sum
$\Phi(\Gamma)$	$\Phi_0 \cdot e^{-(\nu-\nu_0)^2/2\Gamma^2} \cdot S_{26}$	Phonon resonance
R	$F_{U,Bi}/F_U$	Buoyancy ratio

$\S$2 Angular Dependence

The angular modulation follows a Gaussian beam profile:

$$\Delta T(\theta) = \Delta T_{\mathrm{CMB}} \cdot \exp\left(-\frac{\theta^2}{2\sigma_\theta^2}\right)$$

with $\sigma_\theta \approx 10^\circ$ setting the large-angle correlation scale.

$\S$3 Angular Sweep

θ (deg)	$\Delta T(\theta)$ (K)	$\Delta T/T_0$
0.1	Peak (on-axis)	Maximum
1.0	$\sim 0.999\times$ peak	Near-peak
5.0	$\sim 0.88\times$ peak	Moderate damping

θ (deg)	$\Delta T(\theta)$ (K)	$\Delta T/T_0$
10.0	$\sim 0.61\times$ peak	Half-power

$\S$4 Physical Interpretation

The S_{26}^3 cube accounts for three independent contributions: 1. **Spatial:** 26D compressed gravity shell structure 2. **Temporal:** Phonon propagation across 26 layers 3. **Spectral:** Resonance coupling at 1.25 THz

The product $S_{26}^3 \cdot \Phi \cdot (2R-1)$ generates temperature fluctuations whose angular power spectrum (PAPER_1092) matches the Planck 2018 acoustic peak structure.

$\S$5 Comparison to Inflaton-Based ΔT

Mechanism	$\Delta T/T$	Free Parameters	Physical Origin
Slow-roll inflaton	$\sim 10^{-5}$	$V(\phi), \epsilon, \eta$	Quantum vacuum fluctuations
SCm phonon	$\sim 10^{-5}$	Φ_0, Γ, [SSq]	Vacuum buoyancy resonance

The SCm approach reduces the parameter count and provides a directly testable prediction: linewidth Γ at 1.25 THz modulates ΔT.

References

- PAPER_1092: SCm CMB Angular Power Spectrum
- PAPER_1094: CMB Buoyancy Sector Lagrangian
- PAPER_1085: Phonon-Modulated Hubble Parameter
- PAPER_199: $F_{U,Bi,i}$ Taxonomy Part 2

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